IEOR 221

2025 Midterm

October 15, 2025

Due: 5 PM (PDT) October 16, 2025

Please present all your answers to 4 decimal places.

(Note: in the markets price per unit of a security may indeed go beyond 2 decimal places when it is traded with large multiples)

1)
Two corporate bonds, B_A and B_B, (from two different issuers, A and B, issued some time ago) both with notional $1000 have 5 year remaining before they mature. B_A has 5% and B_B has 8% coupon paid annually (at year 1, 2, 3, 4, 5). Both bonds are traded liquidly in the markets. The current risk free discount curve is 4.0% flat, annual compounding.

(1) what is the present value of these bonds under the risk free discount curve?
(2) B_A is traded at $947.7016, B_B is traded at $942.4044, what are their internal yield of return (YTM) in annual compounding?
(3) An equity investor is considering investing in shares of one of the 2 companies. The 2 companies are in the same industry sector and their current and forecasted revenue are similar. Which company should she prefer?

2)
An interest rate swap that exchanges IBOR with fixed rate = 4.5% semiannually has 20 month remaining. Its notional is $100M. The 6mo IBOR fixed 4 month ago is 4.25%. The IBOR forward for 2 to 8, 8 to 14 and 14 to 20 month are 4.75% and 5% and 5.5%. The 2, 8, 14 and 20 month OIS rates (continuous compounding) are 3.75%, 4%, 4.25% and 4.25% respectively. What is the value for the swap holder who pays fixed rate? Ignore counter party credit risk. Provide your answer in $M.

3)
A client asked 3 financial institutions (A, B and C) to price a 10 year forward contract to buy stock S that currently traded at $100. The company pays yearly dividend (once per year). The last dividend was paid 6 month ago at $8 per share. It is expected to pay the same amount 6 month from now as well. The risk free interest rate is 4.5% (continuous compounding) flat. Institution A priced the forward by assuming the stock will pay $8 dividend every 12 month starting 6 month from now; B priced it by assuming the stock pays 8% continuous dividend yield throughout the life of the forward; Trader at C believes in the next two years the company will pay $8 annual dividend but not sure if the company will adjust dividend thereafter as stock price changes. Furthermore she thinks that continuous dividend yield is not a good approximation. Hence the trader at C priced the forward with the assumption that the stock will pay $8 dividend at 6th and 18th month, and 8% proportional dividend every 12 month thereafter (i.e., at 30th month the

company pays dividend equals to 8% of its then stock price. This payment will be repeated every 12 month before maturity). Assume the continuous dividend yield and the discrete annual proportional dividends are reinvested in the stock.

(1) What is the forward prices institution A gave to the client?
(2) What is the forward prices institution B sent to the client?
(3) What is the forward prices institution C sent to the client? Explain how she calculated it.

4)
A 2 year European call on a non-dividend paying stock with spot 100 and strike 101 is traded 18.5. However there is no 2 year put option at this strike traded in the market. A client asks an investment bank (IB) to write a 2 year European put option with the same strike. The risk free rate is 4% (continuous compounding). Assume frictionless stock and call option markets.
(1) What is the fair price for the put option that the IB should quote to the client?
(2) Devise a static hedging position for the IB so that it is immune from the directional moves of the stock price after it writes the put option according to the quote given in (1).

5)
A non-dividend paying stock is currently traded at 90. The annualized volatility is 35%. The risk free interest rate is 5% (continuous compounding).
(1) What is the value of a 1 year European put option with strike = 92? Price it using a two step CRR binomial tree.
(2) What if the put is American? Price it using a two step CRR binomial tree.